Localization, Grasping, and Transportation of Magnetic Objects by a Team of MAVs in Challenging Desert-Like Environments

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***Abstract*—Autonomous Micro Aerial Vehicles (MAVs) have the potential to assist in real-life tasks involving grasping and transportation, but not before solving several difficult research challenges. In this work, we address the design, control, estimation, and planning problems for cooperative localization, grasping, and transportation of objects in challenging outdoor scenarios. We demonstrate an autonomous team of MAVs able to plan safe trajectories for manipulation of ferrous objects, while guaranteeing interrobot collision avoidance and automatically creating a map of the objects in the environment. Our solution is predominantly distributed, allowing the team to pick and transport ferrous disks to a final destination without collisions. This result is achieved using a new magnetic gripper with a novel feedback approach, enabling the detection of successful grasping. The gripper design and all the components to build a platform are clearly provided as open-source hardware for reuse by the community. Finally, the proposed solution is validated through experimental results, where difficulties include inconsistent wind, uneven terrain, and sandy conditions.**

***Index Terms*—Aerial systems: applications, field robots, swarms.**



Fig. 1. Navigation of three MAVs in the Abu Dhabi desert.

I. Introduction

MICRO Aerial Vehicles (MAVs) equipped with onboard sensors are ideal platforms for autonomous navigation in complex and confined environments for solving tasks such as exploration [1], inspection [2], mapping [3], interaction with the environment [4], [5], and search and rescue [6]. In the recent years, MAVs have been endowed with manipulation and transportation capabilities [7]–[10]. This has contributed to increase their ability to solve tasks involving environment interaction and manipulation, without limiting their operations to surveillance and search and scenarios. In this context, the opportunity to use multiple MAVs can contribute to increase the efficiency of missions, solving multiple tasks in a cooperative manner.

In this work, we address the design, control, planning, and estimation problems to cooperatively localize, acquire, and transport ferrous objects in a challenging outdoor desert scenario with uneven terrain and inconsistent winds as shown in Fig. 1. Teams of MAVs can provide additional benefits compared a single vehicle. For example, it is possible to eliminate single points of failure while achieving exploration and mapping more quickly. Although the complexity of such systems increases with the number of MAVs, the exchange of information between vehicles can be exploited by each agent in order to make better decisions, plan motions, and allocate tasks.

Relevant to this work are results focusing on areas involving multi-MAVs ranging from autonomous navigation to grasping

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for environment interaction. Different works have analyzed the problem of aerial manipulation and transportation in indoor scenarios using cables [8], [11], [12], using rigid attachments [7], and multi-arm manipulator systems [9], [13]. All of these works rely on motion capture systems for state estimation and focus on ensuring a static equilibrium of the payload at a desired pose while respecting constraints on the tension. Other works explore similar tasks leveraging multi-MAVs for transportation of objects using cables, but with larger vehicles and in real outdoor scenarios [14], [15]. In the proposed work, grasping does not rely on cable connections but rather a rigid connection between the MAV and the object. Moreover, the systems were already deployed with connections between the agents. Conversely, our system must handle the additional difficulty of switching between different states ranging from navigation to finding and grasping objects in real-time.

Multi-vehicle coordination and planning problems are analyzed in [16]–[18], where the grasping and transportation problems are not considered. Other approaches analyze the scheduling and dynamic reallocation problems [19], [20]. In all these cases, motions were limited to indoor environments, and the works do not address the object identification and grasping tasks.

This work presents multiple contributions. First, we present how to obtain a reliable state estimation and control for each vehicle, enabling successful navigation and grasping of ferrous objects. Second, we convey the design and coordination to enable a team of aerial robots to identify and collect ferrous objects with the overarching objective to transport them to a final common destination. We also provide an open-hardware gripper design for reuse by the community. The approach introduces a compliance in the grasping mechanism, obtained using an elastic element in the join. This passive compliance is very convenient in grasping and interaction tasks with aerial vehicles as already demonstrated in multiple works [21]–[23]. The proposed system, including gripper and all sensors, is very affordable since it costs less than $3500 USD. Finally, this is the first time that estimation, planning, and control problems are addressed simultaneously in a challenging scenario such as in a desert. Uneven surfaces, inconsistent light conditions, and strong winds make the grasping and navigation problems even more challenging. We provide our insights and solutions to these problems, and we believe that the proposed solution, both in terms of hardware and software, can be useful for search and rescue and delivery scenarios in extreme conditions. In addition, the approach can be adopted by other groups in the community, since most of the hardware components are off-the shelf. A simplified version of the proposed approach (applied in a flat, structured environment, in comparison with the varied terrain of sand dunes presented in this work) exhibited the best performance among 20 participants in the "Treasure Hunt" challenge of the Mohamed Bin Zayed International Robotics Challenge (MBZIRC[1] 2017). This competition attracted wide interest in the robotic community (143 teams registered), was one of the motivations of this research, and served as a valuable and trustworthy benchmark for evaluation of our system's performance compared to state-of-the-art solutions.

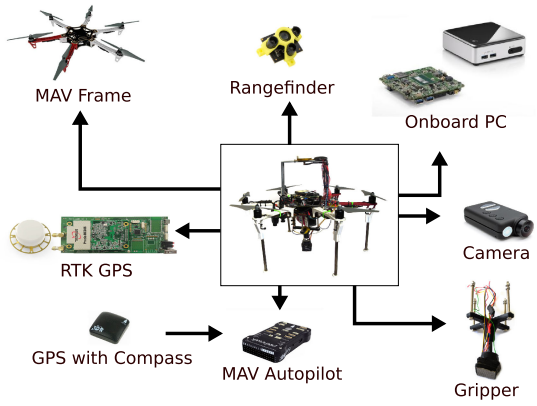


Fig. 2. The proposed system architecture.

The paper is organized as follows. Section II shows the platform design and the grasping mechanism. Section III presents our methodology to generate trajectories for multi-agent navigation while avoiding inter-robot collisions when communication is available, and Section IV discusses the dynamics and control of the vehicle. Section V explains our strategy to obtain the pose of the vehicle at a high rate for autonomous navigation, and it describes the target detection. Section VI presents our experimental results, and Section VII concludes the work and presents future opportunities.

II. Platform Design

The proposed system is depicted in Fig. 2. Each individual platform is composed of a DJI hexacopter frame F550,[2] DJI E310 motors,[3] a gripper, computational resources, and sensor modules. An Intel-NUC-i7[4] is used for on-board computation, a Mobius ActionCam[5] camera is employed for object detection and relative visual localization, and a TeraRanger rangefinder[6] is used for altitude estimation. The system is controlled on the lowest level by a PixHawk autopilot, which contains a set of sensors such as accelerometers, gyroscopes, and magnetometers that are necessary to stabilize the attitude of the MAV. We use a combination of GNSS (Global Navigation Satellite System) attached to the Pixhawk autopilot and Real Time Kinematic (RTK) satellite for localization. The RTK system improves the single GNSS localization module with data derived from satellite-based positioning systems such as GPS, GLONASS, Galileo, and BeiDou. Information on the position is provided in the RTK system by a PRECIS-BX305 GNSS RTK BOARD (GPS L1L2/GLONASS G1/BEIDOU B1B3), providing a final accuracy of 10 millimeters (mm) ± 1 parts-per-million (ppm) horizontally and 15 mm

[1] http://mbzirc.com

[2] https://www.dji.com/flame-wheel-arf/feature

[3] https://www.dji.com/e310

[4] https://www.intel.com/content/www/us/en/products/boards-kits/nuc/kits/nuc7i7bnh-16gb-optane.html

[5] https://www.mobius-actioncam.com/

[6] http://www.teraranger.com/products/teraranger-one/

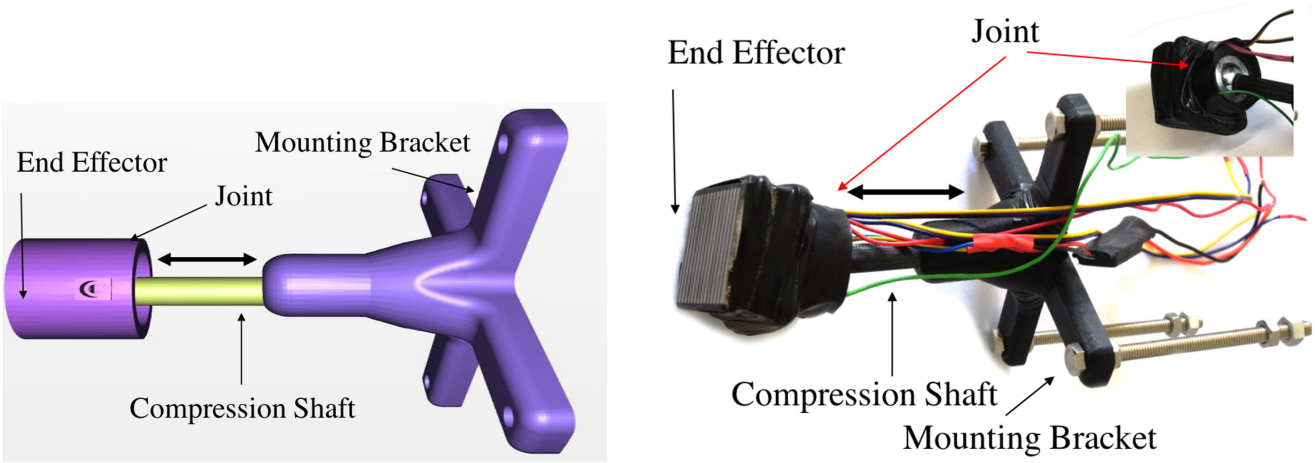


Fig. 3. The designed gripper (left) and its realization (right).

± 1 ppm vertically when the RTK device is in the most accurate state, "RTK FIX". In principle, the vertical position (altitude) provided by the RTK GPS is measured above the mean sea level, which means that the MAV does not have precise information about the ground level profile and the height of objects for grasping, which is supplemented by the rangefinder measurements.

A key component in our design is the gripper, which must be lightweight and able to acquire and carry flat ferrous objects as required by the MBZIRC subchallenge. The gripper consists of 3D printed parts and includes a mounting bracket, a compression shaft, and an end effector as shown in Fig. 3. The mounting bracket connects to the ventral part of the vehicle and provides a receptacle for the compression shaft. Internally, a spring allows the shaft to retract during take off and landing to avoid damaging the gripper, and it also provides axial compliance when grasping. The shaft is connected to the end effector of the gripper via a ball-and-socket joint. The end effector is the core of the gripper; an electro-permanent magnet OpenGrab EPM v3[7] that combines the advantages of electro and permanent magnets and creates a strong magnetic contact with the ferrous object. The payload capacity of the electromagnet is determined by

$$F = m_o g s \tag{1}$$

where F is the required nominal force, m_o the object mass and s is a safety factor, which needs to be properly selected still allow an effective object release, independently of the grasping location on the object surface.

We also must account for the decrease in the available holding force of the magnet, which is perpendicular to the end effector magnet plane, in case its direction is not parallel to the gravity one. It was experimentally verified that a holding force of the ferrite magnet decreases by nearly 85% in shear. Using a maximum rotation of 45° of the magnet results in a decrease of approximately 42% in holding strength. This is then accounted for in the above equation with the safety factor. The total weight of the proposed gripper design is 175 g. The EPM Gripper works by taking a 5 V 1 A power to switch states, and it does not require constant power. Finally, another key component is a spherical joint between the shaft and the EPM, which allows mechanical decoupling of the relative rotation between the platform and the object.

Out of the box, the EPM does not provide any feedback information required to identify a successful grasping. However, this kind of feedback is critical for good performance of the system. Without it, in the case of unsuccessful grasping, the vehicles would use precious battery time flying to the drop location and back, without carrying anything. To address this problem, two Hall effect sensors are installed on the sides of the EPM. When a ferrous object is attached, the magnetic field around the magnet changes. This is reflected by a change in the output voltage of the Hall effect sensors. To prevent false-positive detections, a short calibration is done before each grasping attempt during which 100 samples of the output voltages are averaged and experimentally a new threshold is set above the average. If the measured voltage exceeds the set threshold, a successful grasp is signaled. Likewise, if the object is dropped during flight, the measured voltage drops below the threshold and an object loss is signaled, which allows the MAV to abort the delivery and attempt grasping another object.

A custom board was designed to provide a low-level interface between the main computer, the gripper, and a serial radio module. This board uses an 8-bit ATxmega 128A4U[8] microcontroller and an FT232RL[9] USB serial driver to communicate with the main computer. The gripper is controlled by a PWM signal, and the feedback voltage from the Hall effect sensors is measured by the microcontroller's ADCs. The board is also fitted with a socket to accommodate an XBee Pro radio module, which is used to relay RTK corrections from the base-station to the RTK GPS module. A low-level architecture is shown in Fig. 4. We release the reference files to make and 3D print the

[7]http://nicadrone.com/index.php?id_product=72&controller=product

[8]http://www.microchip.com/wwwproducts/en/ATxmega128A4U

[9]http://www.ftdichip.com/Products/ICs/FT232R.htm

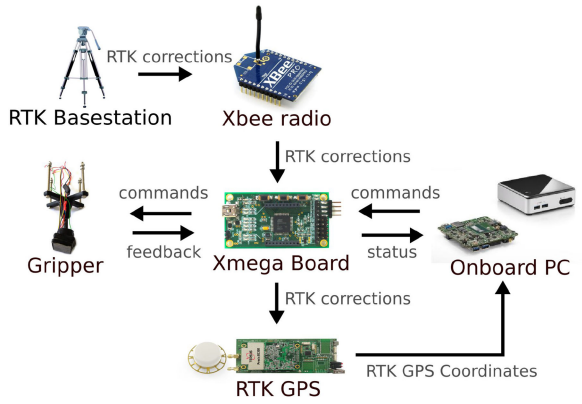


Fig. 4. The low-level interface with communication and feedback.

gripper as well as a self contained code so that anyone can interface the gripper with a μ Controller.[10]

III. Modeling and Control

The position controller uses the estimated state as feedback to follow trajectories given as an output of the high-level trajectory planner. In many previous works, because the attitude dynamics can be assumed to be faster than the dynamics governing the position, linearized controllers are used for both loops [24], [25]. However, we need the system to be capable of large deviations from the hover configuration during operations like fast mapping of objects or for heavy wind compensation, so we use a nonlinear controller. Let us consider an inertial reference frame denoted by $\left[\,\mathbf{e}_1\,,\,\mathbf{e}_2\,,\,\mathbf{e}_3\,\right]$ and a body reference frame centered in the center of mass of the vehicle with an orientation denoted by $\mathbf{R} = \left[\,\mathbf{b}_1\,,\,\mathbf{b}_2\,,\,\mathbf{b}_3\,\right]$ where $\mathbf{R} \in SO(3)$. The dynamic model of the vehicle can be expressed as

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{v},\; m\dot{\mathbf{v}} = f\mathbf{R}\mathbf{e}_3 - mg\mathbf{e}_3, \\ \dot{\mathbf{R}} &= \mathbf{R}\hat{\mathbf{\Omega}},\; \mathbf{J}\dot{\mathbf{\Omega}} + \mathbf{\Omega}\times\mathbf{J}\mathbf{\Omega} = \mathbf{M}, \end{aligned} \tag{2}$$

where $\mathbf{x} \in \mathbb{R}^3$ is the Cartesian position of the vehicle expressed in the inertial frame, $\mathbf{v} \in \mathbb{R}^3$ is the velocity of the vehicle in the inertial frame, $m \in \mathbb{R}$ is the mass, which includes the mass of the object during the grasping phase using the feedback mechanism described in the previous section, $f \in \mathbb{R}$ is the net thrust, $\mathbf{\Omega} \in \mathbb{R}^3$ is the angular velocity in the body-fixed frame, and $\mathbf{J} \in \mathbb{R}^{3\times3}$ is the inertia matrix with respect to the body frame. The hat symbol $\hat{\cdot}$ denotes the skew-symmetry operator according to $\hat{\mathbf{x}}\mathbf{y} = \mathbf{x}\times\mathbf{y}$ for all $\mathbf{x},\mathbf{y} \in \mathbb{R}^3$, g is the standard gravitational acceleration and $\mathbf{e}_3 = \left[\begin{array}{ccc}0 & 0 & 1\end{array}\right]^\top$. The total moment $\mathbf{M} \in \mathbb{R}^3$, with $\mathbf{M} = \left[\begin{array}{ccc}M_1 & M_2 & M_3\end{array}\right]^\top$, along all axes of the body-fixed frame and the thrust $f \in \mathbb{R}$ are control inputs of the plant. The dynamics of rotors and propellers are neglected and it is assumed that the force of each propeller is directly controlled. The total thrust, $f = \sum_{j=1}^{6} f_j$, acts in the direction of the z axis of the body-fixed frame, which is orthogonal to the plane defined by the centers of the six propellers. The relationship between a single motor thrust f_j, the net thrust f, and the moments $\mathbf{M}$ can be written as

$$\begin{bmatrix} f \\ M_1 \\ M_2 \\ M_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ sd & 1 & sd & -sd & -1 & -sd \\ -cd & 0 & cd & cd & 0 & -cd \\ -1 & 1 & -1 & 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \\ f_6 \end{bmatrix} \tag{3}$$

where $c = \cos(30^\circ)$, $s = \sin(30^\circ)$ and d is the distance from the center of mass to the center of each rotor in the $\mathbf{b}_1, \mathbf{b}_2$ plane. For non-zero values of d, (3) can be inverted using the right pseudo-inverse.

For control, we build on the work in [26] and [27] with control inputs $f \in \mathbb{R}$ and $\mathbf{M} \in \mathbb{R}^3$ chosen as

$$\begin{aligned} f = -\Bigg(&-k_x\mathbf{e}_x - k_{ib}\mathbf{R}\int_0^t \mathbf{R}(\tau)^\top\mathbf{e}_x d\tau - k_{iw}\int_0^t \mathbf{e}_x d\tau \\ &- k_v\mathbf{e}_v - mg\mathbf{e}_3 + m\ddot{\mathbf{x}}_d\Bigg)\cdot\mathbf{R}\mathbf{e}_3, \end{aligned} \tag{4}$$

$$\begin{aligned} \mathbf{M} = &-k_R\mathbf{e}_R - k_\Omega\mathbf{e}_\Omega + \mathbf{\Omega}\times J\mathbf{\Omega} \\ &- J\left(\hat{\mathbf{\Omega}}R^\top R_c\mathbf{\Omega}_c - \mathbf{R}^\top\mathbf{R}_c\dot{\mathbf{\Omega}}_c\right), \end{aligned} \tag{5}$$

with $\ddot{\mathbf{x}}_d$ the desired acceleration, k_x, k_{ib}, k_v, k_R, k_Ω positive definite terms. We extend the referenced controllers by including two integral terms, which accumulate error in the body frame and the world frame, respectively. In another work [28], an integral action is considered in the world frame. These terms are extremely important considering the operating conditions of strong and inconsistent wind, for example around dunes in the desert. Both terms provide the opportunity to capture external disturbances (e.g., wind) separately from internal disturbances (e.g., an inefficient prop or a payload imbalance), particularly when the vehicle is permitted to yaw or rotate about the vertical axis. The thrust and moments are then converted to motor rates according to the characteristic of the proposed vehicle. The subscript C denotes a commanded value and $\mathbf{R}_C = \left[\,\mathbf{b}_{1,C}\,,\quad \mathbf{b}_{2,C}\,,\quad \mathbf{b}_{3,C}\,\right]$ is calculated as

$$\begin{aligned} \mathbf{b}_{2,des} &= \left[\,-\sin\psi_{des},\quad \cos\psi_{des},\quad 0\,\right]^\top,\; \mathbf{b}_{3,C} = \tfrac{\mathbf{f}}{||\mathbf{f}||}, \\ \mathbf{b}_{1,C} &= \tfrac{\mathbf{b}_{2,des}\times\mathbf{b}_3}{||\mathbf{b}_{2,des}\times\mathbf{b}_3||},\; \mathbf{b}_{2,C} = \mathbf{b}_3\times\mathbf{b}_1,\; \hat{\mathbf{\Omega}}_C = \mathbf{R}_C^\top\dot{\mathbf{R}}_C, \end{aligned} \tag{6}$$

where $\dot{\mathbf{R}}_C$, is obtained differentiating with respect to time the vector elements composing $\mathbf{R}_C$. Note that here we have to define $\mathbf{b}_{2,des}$ based on the yaw instead of defining $\mathbf{b}_{1,des}$ as done in [27] due to a different Euler angle convention (we use the ZYX convention instead of ZXY). The error quantities $\mathbf{e}_R, \mathbf{e}_\Omega, \mathbf{e}_x, \mathbf{e}_v$ are defined in [26], and they represent the orientation, angular rate errors, and translation errors respectively. The symbol $.^\vee$ represent the *vee* map $\mathfrak{so}(3) \to \mathbb{R}^3$. The exponential stability has been proved for a similar PID controller in [28].

[10] https://github.com/loiannog/MBZIRCgripper

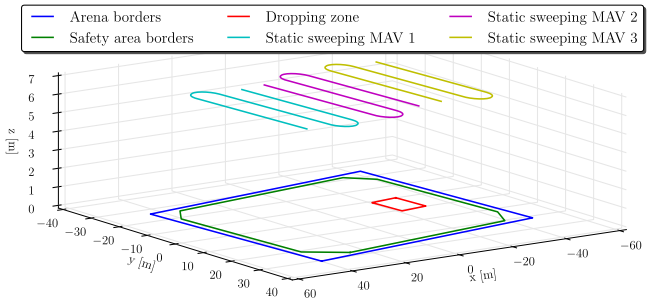


Fig. 5. Coverage trajectories based on Boustrophedon coverage with a Dubins vehicle [32] and area decomposition into three distinct zones, one for each MAV.

IV. Trajectory Planning

Given a pre-defined rectangular area to scan by a team of MAVs, the task first involves generating collision free trajectories for each MAV. Secondly, we must avoid inter-MAV collisions when grasped objects have to be dropped in a common assigned box. A common way for area coverage is to use a Zamboni pattern or a lawn mover pattern. Path planning for such a problem is described as Coverage Path Planning (CPP) [29], where for a given area the CPP provides a path from which the entire workplace can be scanned with an onboard camera. The work in [30] combines graph search techniques and spline-based methods to provide a solution for multi-robot coverage with requirements for sensor resolution and field-of-view.

Instead of solving a combined multi-robot path planning coverage problem, we choose a simple area decomposition method and then assign an independent zone for each MAV to localize and collect objects. The required coverage area is split into equally-sized zones and guarantees that no paths intersect during the coverage phase. Each MAV plans the coverage path using the Boustrophedon coverage [31] in its zone as shown in Fig. 5. A reduced field of view is used for the planning and is set based on the required overlap in the coverage (set to 20%) and on the real FOV camera projection to the ground plane with respect to the sweeping altitude.

While following the trajectory, the MAV detects and localizes the position of any object. Upon a detection, the trajectory tracking is stopped and the MAV tries to grasp the object immediately. After either a successful grasping and dropping of the object or after a number of unsuccessful grasp attempts, the MAV continues sweeping from the last point on the trajectory.

The coverage trajectories are collision free since the robots are in independent zones. However, a MAV may need to pass through zones belonging to other robots in order to drop object after its successful grasping and as well in order to return back into its assigned zone. During this maneuvers, we take a proactive approach to ensure that a collision will not occur. There are many challenges to develop such an onboard collision free planner. Our planner uses a simple idea with different altitudes for each MAV. However, while completing the given task, the MAVs cannot stay only in these altitudes during the mission, as they have to descend for events such as grasping and consequent dropping of the objects. The position of the MAV in x-axis and y-axis does not alter rapidly in these events because they are realized by following strictly vertical trajectories. Therefore, the MAV after leaving its assigned altitude is assumed to be a static obstacle for other MAVs and the finding the shortest free path between the starting and goal points may be easily realized by a visibility graph method [33].

This strategy works due to sharing the estimated positions of MAVs between robots. In the case of a communication dropout, typically occurring during deployment of the multi robot system in real world conditions, using a different altitude for each MAV helps to minimize the likelihood of a collision. Furthermore, since only one dropping zone is available, the MAVs must coordinate themselves so that the dropping zone is only occupied by one MAV at a time. When there is a communication dropout, a simple time-window based approach is used to resolves this issue.

V. State Estimation

The state estimation algorithm must identify the state of each robot and the location of the objects to be grasped.

A. State Estimation for Control

As shown in Section II, the platform is equipped with several different sensors such as GPS, a height sensor, and cameras. These sensors can be fused using the complementary nature of each sensor in terms of accuracy and speed to obtain a reliable state estimate to control the MAV. The Extended Kalman Filter available on the PixHawk fuses the inertial sensors, altitude pressure sensor, and the GPS receiver. However, for the considered task, the precision and accuracy for grasping is extremely important since the main goal is to be able to transport the objects to a final destination. The position estimate in the lateral axes is based on the estimate provided by PixHawk, namely positions $\mathbf{x}^p$, and velocities $\dot{\mathbf{x}}^p$. Although, its precision might be satisfactory locally, it is prone to heavy drift in short time ranges. This drift is corrected by differential RTK GPS to ensure repeatability of experiments. Position measurements from the RTK GPS receiver are fused using a Linear Kalman Filter with model

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} \Delta t \\ \Delta t \end{pmatrix}, \tag{7}$$

where $\mathbf{x}^e_{[n]} = \mathbf{A}\mathbf{x}^e_{[n-1]} + \mathbf{B}\mathbf{u}_{[n-1]}$ is the linear system equation, $\mathbf{x}^e_{[n]} = (x, y)^\top_{[n]}$ is the state vector finally used for control, and $\mathbf{u}_{[n]}$ is the system input. The input vector $\mathbf{u}$ consists of velocities obtained by differentiating positions $\mathbf{x}^p$, which ensures that our filter does not introduce any more drift to the resulting estimate when no RTK GPS corrections are involved due to inaccuracies of $\dot{\mathbf{x}}^p$. In situations when the position is not being corrected, the resulting estimate follows the same relative state trajectory as $\mathbf{x}^p$, just shifted according to the last correction.

The multi-robot scenario requires a common reference frame between all vehicles. The base of our Cartesian system is set to predefined GPS coordinates, and its orientation is according to the *ENU* convention, thus the 1st, 2nd and 3rd axis points to the East, North and Upwards respectively. This is the

convention used in Section III to define the inertial frame. A point of origin is measured using the RTK GPS, to which all independent coordinate systems of all MAVs are then shifted after each of them is powered up. The common base station of the differential RTK GPS then ensures that all MAVs estimates are corrected to lie within the same global coordinates. In this way, the estimation of the altitude does not only rely on the pressure sensor, which is not accurate. A Linear Kalman Filter is used to incorporate altitude corrections from the differential RTK GPS, the downward-facing TeraRanger height sensor, and the object detector, which is able to provide an estimate of the relative distance when flying above an object. The estimator provides an option to switch between those sources of data depending on the current task and the circumstances. Correcting the altitude using the TeraRanger rangefinder is feasible when flying above uneven ground, but it cannot be reliably used when the downward-facing sensor is obstructed e.g., when carrying an object, or when there might be a foreign object on the ground. On the other hand, the RTK GPS is able to provide precise relative altitude measurements, but only when RTK FIX has been established, which depends on GNSS signal strength and the quality of the communication link between the base station and the MAV. Further, it is unaware of the ground profile, so it cannot directly provide an estimate of the height above the ground. Finally, correcting the altitude using data from the *object detector* might bring in unexpected steps in the signal due to false positive detections. To be robust to unreliable measurements, which can compromise the mission in such critical settings, a safety mechanism for detecting anomalies is used which can toggle off any of the altitude sensors from being fused.

B. Object Detection and Localization

As mentioned in Section II, the Mobius camera is used to detect the colored, round objects. This high-resolution camera has a rolling shutter, so vibrations induced by the drone motors can cause noisy effects in the image which makes the use of geometry-based methods for object detection (e.g., Hough transform) problematic [34]. Thus, we designed a computationally efficient ellipse detection algorithm, which relies on the use of statistics that are robust to this type of noise [35]. The experiments have shown that a better object detection performance is achieved with a high-resolution rolling shutter camera than with a lower-resolution, global shutter camera of similar price.

Unlike the original method [35], which used adaptive thresholding to detect black-and-white patterns, our pixel labeling was based on a 3D RGB look-up grid, which is either pre-loaded or build semi-automatically during a calibration procedure. During calibration, the MAV hovers (or is held by hand) over the objects in a known spatial configuration and displays the results of the detection on a GUI. The system operator can indicate the object positions and false detections, creating and refining a Hue-Saturation-Value Gaussian mixture model of the object and background colors. Once the user indicates that the detection performance is satisfactory, each cell of the RGB look-up grid (corresponding to the given color) is classified as an object or background using the Gaussian mixture model, and the look-up grid is saved. The main advantage of the RGB look-up grid is that it can implement any pixel-wise classification method in a very efficient way – classification of a given pixel is performed simply by retrieving its class from the cell that corresponds to the pixel's RGB values.

The detection algorithm is based on [35], which searches for continuous segments of object-colored pixels, establishes their bounding box, number of pixels, centroid, convexity, and compactness, and uses these statistics to reject segments that cannot correspond to circular objects. Finally, using the known object size, elliptical shape and camera parameters, the method calculates the relative 3D position of the object which is transformed into the inertial coordinate frame. If the detected object's position and the ground distance obtained from the robot's state estimate are not within a threshold, then the object is marked as a false positive. For details on the method, please refer to [36]. Targets that are trying to be grasped are filtered out in other MAVs to prevent simultaneous grasping of the same target. The following empirical laws are adopted during the manipulation task

- Objects which have not been seen for more than 5 seconds are *deactivated*.
- Objects which are deactivated for more than 3 seconds are deleted from the map.
- Measurements from the *object detector* are paired with objects in the map using a min-distance bipartite graph matching, constrained by the color of the objects.
- Objects located outside of the working area are deleted from the map, and new measurements in such areas are discarded.

Additionally, the grasping attempts might not be successful at all times. The approach also allows a temporary ban for 30 s in a 4 m radius around a particular object to avoid deadlock in the grasping state machine.

VI. Experimental Results

In this section, we present results obtained in the demanding environment of the desert of Abu Dhabi in the United Arab Emirates as shown in Fig. 6. The key objective of this work is to have a team of MAVs identify, grasp, and deliver a number of objects in an environment with an unknown ground profile, difficult wind, and unreliable communication.

The entire pipeline has been implemented in ROS, which is also responsible for the management of communication between MAVs. ROS by design is not suitable for distributed implementation, so we make use of the ROS multimaster package [37] that allows managing nodes, topics, services, parameters across different robots. To reduce load in the communication channels, only selected information (topics) are shared with other team members such as the MAV position in the global coordinate system, the current state of the high-level state machine, the position of the object during grasping, and the planned trajectory. This information is used in nodes for collision free planning, fail-safe reactive collision avoidance, and object estimation.

The proposed approach has been tested with a set of 2 and 3 MAVs as shown in Fig. 7 on the left and right, respectively. The



Fig. 6. One of the vehicles during a grasping task in the desert (on the left) and two vehicles during the navigation task after grasped one object, respectively (on the right), which are circled in green.

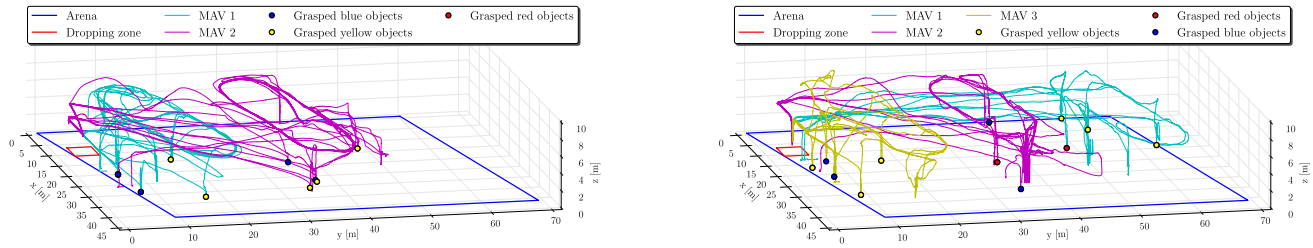
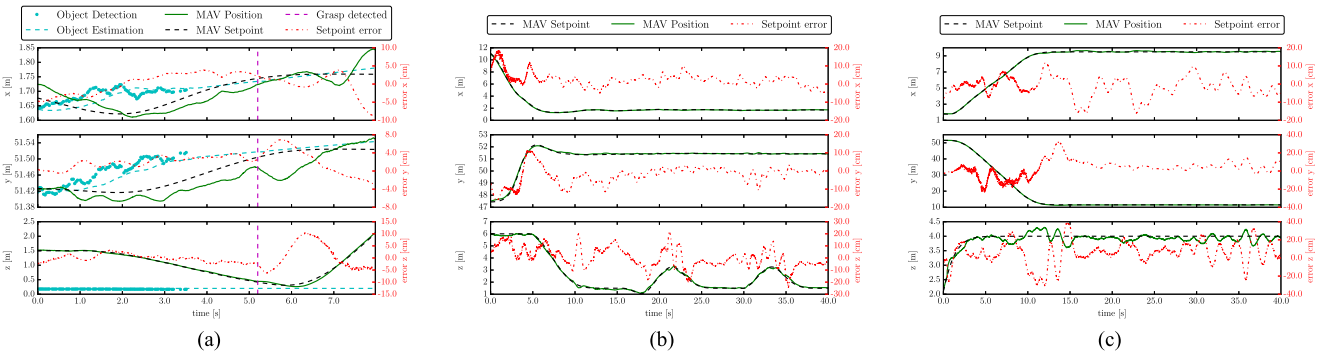


Fig. 7. Path for different vehicles during the grasping task of two different (on the left) and three different types of objects (on the right).



(a) (b) (c)

Fig. 8. Grasping per cartesian axis with object estimation and MAV position, setpoint, and setpoint errors. (a) During grasping attempt. (b) Before grasping attempt. (c) After successful grasping.

weight of each object is 70 g and is placed close to the ground at different altitudes (see Fig. 6 right indicated with green ellipses) in a total area of $70 \times 45\,\mathrm{m}^2$. Each robot travels an average of 250 m in the x and y Cartesian axes at an average height of 10 m. In Fig. 7, each object is identified by the color corresponding to the vehicle that performed the grasping. We can clearly identify a total of 12 objects. Each vehicle picked up and delivered at least 3 objects thus showing robustness of the system across platforms. The dropping area is identified in the top left corner of Fig. 7 by a red square on the ground. During the experiments, the robots experienced wind speeds up to 10 m/s. Regardless, over the course of 10 trials, the system achieved an average success rate of 90% when grasping and 95% when dropping objects in the goal location.

We finally discuss the control performance during the navigation and grasping task. In Fig. 8, we report on the results for the grasping task executed by one of the platform. The vehicle starts at 1.5 m above the object. During the descent phase, which starts at 1 s, the vehicle tries to center with respect to the object on the other two Cartesian axes. The control errors are, on all axes, within few centimetres, which is sufficient to successfully grasp the objects. The attached multimedia material shows many trials with similar results independently of the color and position on the terrain of the objects. In addition, in Table I, we report the average mean tracking errors before and after the grasping task. The errors are equivalent on the same magnitude range, showing that our control strategy is robust with respect to the weight changes introduced by the payload.

TABLE I
AVERAGE POSITION MEAN ERRORS IN METERS DURING NAVIGATION BEFORE AND AFTER THE GRASPING TASK

Average Error	Before Grasping (m)	After Grasping (m)
x	0.032	0.045
y	0.038	0.071
z	0.067	0.110

VII. CONCLUSION

In this work, we developed an approach to enable a team of MAVs to identify, grasp, and deliver objects in challenging desert-like environments. The key challenges included creating a robust gripper for ferrous objects, scanning and locating the objects of interest, uneven terrain, inconsistent wind, and the potential for inter-robot collisions, all of which make the proposed task extremely difficult. We showed the effectiveness of our approach in real tests with the ability to have multiple vehicles simultaneously collecting different colored ferrous objects in the desert. We release the main platform hardware components and designs as open source, which is useful for other researchers to create an autonomous flying robots. We believe that the proposed solution, both in terms of hardware and algorithms, will be useful in industrial automation and search and rescue scenarios. Future strategies will investigate how to carry larger magnetic objects with multiple vehicles in a cooperative manner. We will also explore improved state estimation approaches, eliminating the need for GPS, which would open up opportunities for GPS-denied tasks.

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